SecFob: A Remote Keyless Entry Security Solution

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***Abstract*—All major vendors currently offer Remote Keyless Entry (RKE) systems for passenger vehicles, which have become a standard feature of modern cars. These systems enable wireless communication between drivers and vehicles, facilitating convenient functions such as remote start and passive keyless entry. Unfortunately, the existing security measures implemented in these systems are often inadequate, leaving them vulnerable to intrusions and a wide array of attacks. To address these vulnerabilities, we present SecFob, a solution that combines quantum random number generation (QRNG), machine learning, and encryption techniques. SecFob is designed to defend against common attack techniques by leveraging the perfect entropy offered by QRNG and by incorporating machine learning techniques, specifically utilizing the Siamese Neural Network. These solutions are cost-effective, portable, and with the use of ML technique, are able to detect amplified signals.**

***Index Terms*—Remote Keyless Entry, Quantum Random Number Generator, Machine Learning, Radio Frequency.**

I. Introduction

In modern vehicle advancements, we are seeing many innovations in designs, performance, and features to make vehicles of all types more enjoyable and user-friendly. These advancements, including but not limited to automated vehicles, bring massive cyber security problems into light [1]. All modern vehicles have one major security-risk area: the *key fob* and *Remote Keyless Entry* (RKE) system. The key fob allows wireless communication between the driver (possessing the key fob) and the vehicle to perform various functions. Newer RKE systems include remote start, trunk opening, and passive keyless entry features. These systems are gradually completely removing the physical car key that was inconvenient and vulnerable to key copying and lock picking. Despite the popularity of these systems, very little security has been applied to the area, making them vulnerable to intrusion and a wide variety of attacks [2]. Without security measures in place, intruders can gain unauthorized access to a vehicle and its contents.

Research on attacking key fob security vulnerabilities started almost two decades back [3]. Over time, various attacks have been demonstrated, which include relay attacks, scan attacks, dictionary attacks, and two-thief attacks [3]–[6]. To keep the vehicle secure against such attacks, various schemes [7], [8] have also been proposed and have shown promising defensive capabilities. However, the main issues with the proposed solutions are their complexity, affordability, and ability to defend against large number of attack vectors. One of the hardest-to-defend attacks to date is the two-thief attack. As the name suggests, this attack involves two attackers, one close to the key fob and another close to the vehicle. The one close to the key fob captures the signal, amplifies it, and sends it to the one close to the vehicle. This signal is replayed forcing the vehicle into thinking the real key fob is in the vicinity and sending the commands. This clever attack method is extremely effective due to the current encryption and authentication methods being bypassed.

In this paper, we propose, SecFob, a novel solution that effectively defends against each of the common attack methods using a combination of Quantum random number generator (QRNG), machine learning, and encryption innovation. Quantum-based random number generation has evolved over recent decades, making more affordable and portable options available [9]. For example, quantum chip technology has been used in cell phones to create a much higher entropy security system for access [10]. Machine learning and artificial intelligence have been booming topics in technological innovations and advancements, which we take advantage of to automate the detection of amplified signals (basis of some of the popular attacks, like the two thief attack mentioned above) in SecFob. More specifically, SecFob uses a machine learning framework called Siamese Neural Network, which compares two images and reports their similarity levels [11]. It has been successfully used in facial recognition and handwritten signature forgery screening scenarios with promising results.

II. Background and Related Work

A. Keyless Entry Architecture

Vehicle keyless entry systems rely on the key fob programmed by the manufacturer and work with the involvement of the vehicle Body Control Module (BCM) and Electronic Control Unit (ECU). The key fob sends radio frequency (RF) packets to the car as a command to perform functions such as unlock, lock, trunk open, hood open, remote start, and often more. The key fob sends the command as a repeating signal to ensure the vehicle will receive one of them [3]. For example, if the key fob unlocks button is pressed, the fob will send a total of four unlock commands in one packet to have redundancy in place. This ensures the vehicle intercepts at least one unlock signal excerpt and follows through with the given command. The key fob is powered by a small cell battery within its small frame, a transmitter, and a small circuit board programmed to communicate with the specific vehicle it is programmed to. The microcontroller used in the key fobs

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is inexpensive and of limited capacity, making the range of the signal about 20 meters [3]. The key fob will contain a proximity-based immobilizer chip that must be within a certain distance threshold for the vehicle to approve the actions.

B. Remote Keyless Entry Attack Methods

Various attacks against vehicles' keyless entry systems have been applied and studied. The success of these attacks is primarily due to the intruder's unlimited access to the vehicle's door handle and their capability to extract information from the owner's key fob by generating low-frequency signals near the owner [6].

Dictionary Attack– is when a collection of codes or commands are transmitted to the vehicle until one works [3]. The attacker can continuously pull on the vehicle door handle to change the current code and parameters and hopefully grant access to the vehicle.
Scan Attack– this attack uses the *rolling code* technique. The attacker sends different codes to the vehicle transceiver until a match is found with the one in the vehicle [3]. This is the most direct attack from the attacker's point of view since it does not need any other technical information about the system.
Replay Attack– or man-in-the-middle (MITM) attack is when an attacker records a command from the target key fob to use later and to gain unauthorized access to the vehicle [3]. Replay attacks can be performed by a single attacker with a transceiver, making it affordable and discrete.
Reverse Engineering Attack– or challenge-forward prediction attack is performed by a highly skilled individual, usually a mathematician, who records multiple challenges by pulling the door handle and recording the vehicle responses [6]. The challenges are analyzed and mathematically reverse-engineered to predict the next key fob signal in future commands correctly.
Two-Thief Attack– The two-Thief attack is the most successful due to vehicle defenses' inability to differentiate the received signals from its legitimate key fob signals [6]. The Two-Thief attack involves two attackers that aim to create an electronic bridge between the key fob and the vehicle. Thief one starts by triggering the vehicle's door handle and records the vehicle's response. Thief two will transmit the signal to the key fob and record its response. The response is then sent back to thief one to playback to the vehicle for access.
Jamming Attack– this attack occurs when the attacker interrupts the key fob transmission to the vehicle and stops the command from reaching the vehicle [3].

C. Authentication Methods

Three main authentication techniques are used to secure fob-vehicle communication.
Fixed Code Technique: In fixed code systems, a predetermined code is stored on the key fob. If the key fob is pressed and within range of the vehicle, the key sends this code to the vehicle and is checked for validity [3]. If the code is processed as valid, the vehicle will perform the given command. The predetermined code does not change with each toggle of the key fob; it will always stay the same. Attacks such as replay, scan, and dictionary attacks are effective against systems using the fixed code technique. These attacks are successful due to no change between each key fob transmission.
Rolling Code Technique: The rolling code is a system in which the transmitter and receiver of RF signals are synchronized, and the code for each transmission is changed each time it is toggled [3]. The rolling code changes its sequence counter each time the frequency is sent to ensure the next signal will differ from the last. The reverse engineering, two-thief, and jamming attacks can each exploit the rolling code technique by impersonating, mathematically reversing, and interrupting communications with the RKE signal processing procedure. The rolling code technique is outdated and does not contain enough randomness to defend the RKE system effectively.
Challenge-Response Technique: This technique demonstrates how the vehicle can identify if the key fob sending a command is legitimate or not [6]. The vehicle and key fob share a secret encryption key. When the key fob command is toggled, the vehicle sends a random number to the fob. The fob encrypts this number using its encryption key. The encrypted random number is then returned to the vehicle for decryption and validity. If the random number is valid, the vehicle will respond to the original command.

D. Related Work

Considering the limitations of existing authentication techniques used in modern vehicles and the success of various attacks against them, many researchers have developed advanced defense mechanisms.

Hussain *et al.* [13] have suggested TIGHT, a solution where lightweight hash algorithms authenticate the sending and receiving of RKE signals. With this approach, the vehicle unlocks only if the authentication passes. The most effective algorithm to defend against the challenge-response attack turned out to be the SHA-256 algorithm. This approach can secure the system against replay, dictionary, and MITM attacks.

Glocker *et al.* [14] proposed a protocol using a compact symmetric encryption algorithm that was deployed to authenticate the messages sent from the fob to the vehicle. Parameswarath *et al.* [15], on the other hand, suggested using a timestamp and asymmetric cryptography. In their solution, the key fob generates a random value, appends it to the timestamp, then signs the string and sends it to the vehicle. The vehicle can verify the signature because it has the same parameters, i.e., random number generator, seed, and time, and knows the public key. This method prevents replay attacks because it contains a timestamp. However, for this solution to work, the vehicle and the key fob must be synchronized, which is difficult to achieve.

Greene *et al.* [8] use AES encryption and timestamping. Their solution uses time thresholds can help eliminate man-in-the-middle attacks where the signal from the key fob would need to be played back through the intruder transceiver. Another security solution proposed by Aliwa *et al.* [16] suggests

TABLE I: Security Solution Comparison

Application Features	HODOR [7]	LASER [12]	TIGHT [13]	SecFob
Secure against replay/MITM attack	✓	✓	✓	✓
Secure against dictionary attack	✗	✓	✓	✓
Secure against scan and jamming attack	✓	✓	✓	✓
Secure against reverse engineering attack	✓	✗	✗	✓
Trainable With New Data	✓	✗	✗	✓
True Random Nonce	✗	✗	✗	✓

Secret Unknown Cipher (SUC) technique as a clone-resistant structure that creates a low-cost digital alternative to Physically Unclonable Functions (PUF). The SUC acts as a secure anchor for the RKE systems to encrypt and decrypt the signals as they are sent and received by the vehicle. This solution is robust against scan and playback attacks. However, it does not work against the Two-Thief attack as it cannot identify the signal from a faulty source.

Daza *et al.* [12] proposed Laser, a solution where the device sends the hash of the sum of the current time and a secret key for authentication. The communication between the key fob and the device is done through a channel that changes each time and is calculated based on the computed hash value. In their work, the initiating device must computer and constantly send the hash value, which might drain the key fob battery. This method is secure against dictionary, jamming, and relay attack.

Similar to our work, Joo *et al.* [7] propose HODOR, which is developed as an authentication mechanism to be implemented on top of existing keyless entry system authentication methods. This approach uses a fingerprinting technique to detect illegitimate signals. It is based on a classifier created based on a training dataset of legitimate signals, using extensive preprocessing and feature extraction techniques. As a result of this approach, keyless entry systems will be protected against attacks such as replay, jamming, and scanning. The key limitation of the approach is the feature extraction, specific to a vehicle, that is done for the classification model. It is desirable if an ML method can be developed that is independent of such feature extraction steps and more broadly available across vehicles of different makes and models.

SecFob stands out by offering a comprehensive solution that safeguards the fob-vehicle communication against all the attacks detailed in Section II-B. Table I compares SecFob with other state-of-the-art solutions.

III. SecFob System Overview

In this section, we discuss SecFob's system model, the adversarial model, and the cryptographic primitives used in the model. Figure 1 gives an overview of the system and our proposed solution's components.

A. System Model

Our system consists of two parties: a key fob and a vehicle Electronic Control Unit (ECU) (Fig. 1). The manufacturer of the fob-vehicle ECU pair will embed specific information, such as the car identification ID_{car}, secret key K, and threshold t, in their ROM before making the product available to consumers. The communication can either start by pressing the unlock button on the key fob or by touching the car door handle. Three-way communication happens between the key fob and the vehicle if the communication starts by pressing the unlock button. However, two-way communication is triggered if the user touches the door handle.

B. Adversarial Model

We assume the attacker cannot access the key fob physically. Attackers can use large files of signal packets and a transceiver to perform dictionary attacks [12]. We also assume that the adversary can access tools and equipment for their attacks, such as radio-frequency transceivers and small-scale amplifiers. Among these, radio-frequency transceivers are used to communicate with vehicles and intercept signals from key fobs. Small-scale amplifiers connected to the transceivers or deployable amplifiers are used in between the key fob location and vehicle to relay the messages being conveyed.

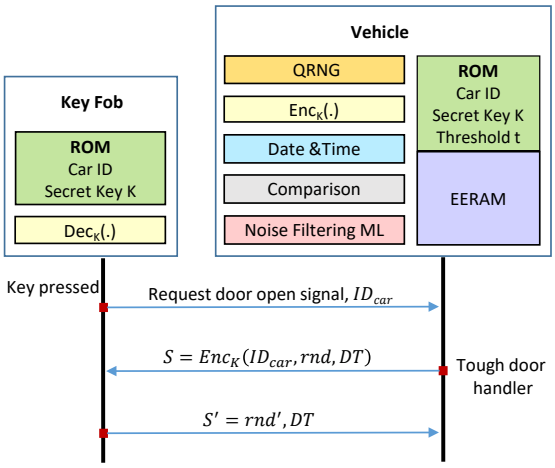

FIG. 1: SecFob System

C. Cryptographic Primitives

To have secure communication, we use the following cryptographic tools.

a) Quantum Random Number Generator.: Quantum random number generators are becoming more prevalent today [10]. The quantum part of the generator works by injecting a laser or other light source into a special crystal that splits

the light source into a random output of photons, which is then converted to random sets. Such entropy generation creates more secure random numbers over the more common software-based pseudo-random generation. Specifically, the resulting number generation here is completely random due to quantum mechanics, making it much more difficult to predict, attack, and reproduce. Luckily, QRNG is being produced in micro size for cell phone systems and is available at an affordable price point, often priced below $10 [9], [10].

b) Symmetric Encryption.: We use symmetric encryption to encrypt the communicated data between the key fob and the vehicle's ECU. Symmetric encryption uses one secret key for both encryption and decryption. The secret key must be shared through a secure channel before use. To comply with the key fob's hardware limitations, our system uses lightweight AES-128 encryption – its implementation uses fewer hardware gates, consumes less power, and has a reasonable throughput [17]. We use $Enc_K(.)$ to indicate symmetric encryption using a secret key K, whereas the corresponding decryption is denoted as Dec_K.

IV. SecFob System

SecFob is a lightweight, secure two-way authentication mechanism triggered by pressing the unlock button or touching the door handle.

A. SecFob Core Principles

At its core, SecFob relies on its true randomness and wide attack array defensive capabilities provided by the QRNG and machine learning. Compared to the conventional pseudo random number generators (PRNG) used in other state-of-the-art defense solutions [7], [12], [13], quantum random number generation provides greater entropy than current PRNGs. With algorithmic advancements in QRNG cryptography optimizations, as seen in [18], QRNGs perform symmetric operations 70 times faster than PRNG equipment and ten times faster when performing asymmetric operations. Additionally, QRNGs can complete these tasks while using 6,000 times less memory, making them lightweight and easier to implement into smaller hardware-based devices [18].

SecFob's second core principle is its artifact filtering with machine learning. Compared to [7], artifact detection and large-scale defensive capabilities are improved in SecFob. Our solution focuses on feature comparisons rather than extraction. The Siamese Neural Network (SNN) can more accurately infer which incoming signals are legitimate or not and even a global model across vehicles of different makes, models, and years is possible. SecFob can detect faulty signals such as those due to amplification easily.

B. General Architecture

As illustrated in Fig. 1, when the unlock button is clicked, the RF signal starts inside the key fob and is sent to the vehicle with the car identifier. Upon receiving the signal, the vehicle verifies the received identity, ID_{car}, by comparing it with the information stored in its memory. If a match is found or the vehicle handle is touched, the vehicle generates a random number rnd, using the QRNG, and obtains the current date and time, DT. The vehicle then encrypts the triple $S = (ID_{car}, rnd, DT)$ using AES encryption with a secret key, K, before sending it back to the key fob. Furthermore, the vehicle's electronic control unit (ECU) stores the triple in its memory (EERAM) for later assessment.

By receiving the triple S, the key fob decrypts and verifies the vehicle's identity before sending the tuple $S' = (rnd, DT)$ back to the car. This step confirms that the signal originates from the vehicle assigned to the key fob – a crucial step when the user initiates communication by touching the car handle. Upon receiving the tuple S', the vehicle's ECU undertakes two actions: a) execution of Algorithm 1, and b) passing of the received signal through the ML inference (Algorithm 2). We explain both the algorithms next.

Algorithm 1 Key Fob Verification Algorithm

```
procedure KEY FOB VERIFICATION(rnd', DT*)
    DT, rnd ← EEROM
    if DT = DT* & rnd = rnd' then
        DT' ← current date time
        if |DT' − DT| ≤ t then
            return valid
    return invalid
```

Algorithm 1 carries out a verification process by cross-checking the received information. This includes comparing the random number, rnd, with the previously stored one in EEROM. Additionally, the algorithm retrieves the current date and time, DT', and examines whether the time difference is within the predetermined threshold, t, set by the manufacturer. Utilizing QRNG for generating random numbers protects against reverse engineering attacks. Additionally, when combined with timestamping, it serves as a means of cross-validation to ensure that the signal was not captured and resent, thereby mitigating risks associated with man-in-the-middle attacks, scan attacks, dictionary attacks, or the use of pre-recorded signals.

Using a trained Siamese Neural Network, the key fob signal is analyzed and classified as approved or not approved. If the signal passes the integrity analysis and Algorithm 1 returns valid, the vehicle ECU will accept the unlock signal and perform the function.If, at any point, the random value does not match, the machine learning notices artifacts in the signal, or the timestamping threshold is not met, the signal will be dismissed, and the unlock function will not be performed.

SecFob ensures that the door will only unlock if both algorithms return a true value, and a failure of either will result in the door remaining locked. This rigorous approach ensures that only legitimate key fobs are granted access to the vehicle.

C. Siamese Neural Network

Machine learning and, more specifically, the Siamese Neural Network are used in our solution to counter an attack such as

Algorithm 2 SNN Algorithm

```
Train SNN model with both amplified and non- amplified keyfob signals.
for each test signal do
    for each amplified signal in train data do
        if Signal is Similar to amplified signal then
            A = Increase Similar Signal Counter
    for each non-amplified signal in train data do
        if  Signal is Different then
            B = Increase Different Signal Counter
if A ≥ B: then
    Test Signal is Predicted as Similar to the Amplified
if B ≥ A: then
    Test Signal is Predicted as Different to Amplified
Compute Accuracy.
Compute test time for each test signal
```

the two-thief attack that uses amplification of a captured signal. We saw physical artifact differences between the normal key fob signals and the amplified ones and chose to use the SNN to discern the signals for our solution. The Siamese Neural Network (SNN) is a class of neural networks that contains two *identical* sub-networks [11]. Here, identical means that each sub-network includes the same parameters and weights. When information is given to the SNN, one sub-network will be compared to the other, and the output informs us of any differences between the signals. One common application of the SNN is in comparison of two pictures – the SNN can focus on small artifacts and details to determine if the two images are highly similar or different. As such, SNN has proven to be an effective tool in facial recognition and signature verification environments [11]. Our goal with the Siamese Neural Network is to feed the machine learning system a new signal and analyze the signal for any amplification or signal artifacts. Moreover, we intend to achieve this identification without any feature extraction method, since those are unlikely to be generalizable across devices. Analysis using SecFob will prevent any possible replayed or forged signals from being accepted to the vehicle.

V. Framework Analysis

MITM attacks are challenging to defend against due to attackers' techniques to capture, possibly from a distance, actual signals and use them with amplification [3], [6]. The intruder is capturing legitimate signals and playing them back immediately or in the future; either way, the vehicle will likely accept the attacker's signal as long as the key fob has not been pressed after the signal was captured. SecFob defends against the MITM attack using two techniques: SNN and timestamp. SNN assists in detecting amplified signals, while timestamp disables the attacker from using the current information in the future. More specifically, SecFob uses the SNN to measure if the incoming signal is amplified or weak and can distinguish if the incoming signal is within certain analog frequency thresholds.

SecFob sets a time-based threshold for the communication process between the two devices. If the command sent from the key fob to the vehicle does not reach within the threshold time t (mostly set to 10 ms), SecFob will not perform the unlock command. The time threshold will not accept any incoming signals outside the threshold, making the jamming attack impossible to achieve due to the signal not reaching the vehicle until after the threshold since the jamming attack involved playing a signal back.

In the two-thief attack, the RF being sent and received between the vehicle and key fob will be amplified in some way to reach the vehicle [6]. If not, the owner would be close enough to either see their vehicle or see the attackers in its close proximity. SecFob defends this attack the same way it defends the MITM attack; by using SNN. The signals are filtered through the SecFob SNN, and the amplified signals are thrown out instantly and not allowed to complete the command. The amplified signals will contain larger RF peaks which SecFob can detect and prevent from accessing.

SecFob defends against the dictionary and scan attacks using the Quantum Random Number Generator. The random number generated by the QRNG changes each time a communication is established. The chances of an attacker being able to predict which code will work successfully would take hundreds of years, given the entropy levels and true randomness of the QRNG [10]. Certain vehicle computers also have input cooldowns where the vehicle will force a 7-second cool down if too many unsuccessful attempts are registered to make the dictionary attack longer and more difficult to complete.

The reverse engineering attack, or challenge-forward prediction attack, is difficult to perform due to its high complexity and skill set required to reverse engineer the algorithm mathematically. Unfortunately, the rolling code has been broken and reverse-engineered [6]. SecFob steps up the security using a QRNG. SecFob's random number generation is exponentially more secure than rolling code and will not let the command through and be accepted into the system.

VI. Experimental Analysis

A. Experiment Setup

The Keyless Entry Systems experiments were on five vehicles with different manufacturing years, brands, countries of origin, and models. The cars were: 2007 Chevrolet Avalanche, 2011 Volkswagen Jetta, 2014 Chevrolet Malibu, 2015 Volkswagen Jetta, and 2016 Honda Accord. For the vehicles manufactured in the United States, the key fob communication was present in the 315 MHz band, while the others were in the 433 MHz band.

B. Radio Frequency Processing

Two software-defined radios were used to capture and playback the key fob excerpts for the analysis of the key fob signals and recording of frequency samples. HackRF

One SDR transceivers [19] were used in tandem to not only capture the signals and replay them but to simulate MITM attacks. The HackRF transceivers were connected to a laptop and desktop computers, each running Universal Radio Hacker (URH) software [20] to record, visualize, and playback the captured key fob signals. To create the amplified key fob signals for our experiments, a wide band high, performance, the low-noise amplifier was installed onto one of the HackRF One transceivers and was used as the dedicated frequency-sending device. Since the amplifier is connected directly to the HackRF One antenna, we can increase the gain of the radio frequency sample in URH and send the amplified signal through the Hack RF One.

C. Training and Testing Data

During the experimentation phase, 15 key fob signals were recorded for each of the five vehicles and stored in the URH software for testing. We sent the key fob signals through HackRF to create the amplified signals. One with the amplifier attached and gain settings increased through one computer while the other HackRF One received and recorded the newly amplified signals. The URH software can display the recorded key fob signal in its analog form, allowing us to see the physical radio waves of each key fob. The key fob signals were recorded at varying distances from the SDR, ranging from one to ten feet, to change the physical structure slightly between each recording. Once the fob signals were recorded into the URH software, we converted the analog signal samples to picture formats for use in the Siamese Neural Network.

D. SNN Use Case

While analyzing the signals in the URH software, we could distinguish the amplified and non-amplified signals. For the vehicles and signal captures described above, the amplified signals were created using the amplifier on one transceiver and sent to the other transceiver for collection. Once the non-amplified and amplified signals were recorded, each signal was converted to a .png format for use in the SNN experiments.

We have leveraged the cloud platform, Google Colaboratory [21], known as Colab, to perform inference operations using SNN [22]. Colab allows anyone to write arbitrary Python code to execute in the browser. Specifically, Colab provides online notebooks that will enable the user to run the Python code without the requirements of the setup of the notebook software. In addition, Colab provides powerful hardware options such as GPUs and TPUs. Machine learning and data science researchers typically leverage Colab to implement inference models. Each of the following experiments was set up using our custom code optimized to work with a GPU to increase speeds of running times.

Each of these tests took anywhere from 12-16 minutes for the ”training” part of the code to run, while the actual execution times for test data analysis were about 3-5 seconds, as seen in Fig 2. This execution time is relative to when you would be waiting for the security solution to run after you click a command on your key fob.

The following experiments were done using the basic idea of five-fold cross-validation and signals from five vehicles. Table II shows the results from all experiments. After receiving high-accuracy results from the Siamese Neural Network, an attack such as the Two-Thief attack will be near impossible to perform.

Same Vehicle, Same Key Fob (SVSK): The SNN algorithm was trained with ten non-amplified and amplified key fob signals and was tested against ten non-amplified and amplified key fob signals from the same vehicle. This was done individually for each of the five vehicles. The test ensured that the SNN could discern between the non-amplified and amplified signals given one car at a time. The average accuracy of these five tests is 90 percent.

Same Vehicle, Different Key Fob (SVDK): This test was to see if the SNN could find any differences between two different key fobs for the same vehicle. The training data was compared against ten more non-amplified and amplified key fob signals from the same vehicle but with the second key fob. Overall accuracy is 90 percent.

Cross Vehicle Evaluation (CVE): The SNN algorithm was trained on a mixture of 40 non-amplified and amplified signals from 4 of the five vehicles. The fifth vehicle not included in the training data was considered the test data and had ten non-amplified and amplified signals. This test was run five times, exchanging the test vehicle out and ensuring the test vehicle data was not in the training data. The average accuracy of these five tests is 92 percent.

Single Global Model: The SNN algorithm was trained on a mixture of 20 non-amplified and amplified, four signals per vehicle. The model was tested against five signals, one from each vehicle. This was rotated five times to test against the various signals. Overall accuracy is 98 percent (see also Table II).

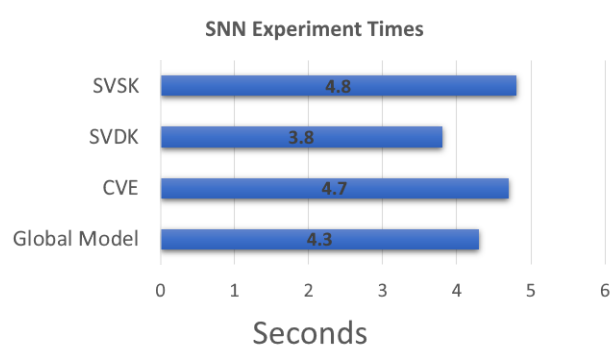

FIG. 2: SNN Execution

VII. FUTURE RESEARCH

With the world of automobiles constantly changing and new features being added every year, the security aspect of vehicle communication needs to be considered. While self-driving and autonomous vehicles are being introduced, Key

TABLE II: SNN Experiment Result Accuracy

Experiments	Vehicle 1	Vehicle 2	Vehicle 3	Vehicle 4	Vehicle 5	Average
SVSK	0.80	0.90	1.00	1.00	0.90	0.92
SVDK	0.70	0.80	1.00	1.00	1.00	0.90
CVE	0.90	1.00	0.80	1.00	0.80	0.90
Global Model	1.00	1.00	1.00	0.90	1.00	0.98

Fobs and keyless entry systems are still the most common form of communication with the car. SecFob uses many different security areas to secure various attack vectors. With advancements in Machine Learning and AI, new RF analysis methods could be substituted with the SNN, or optimizations can be made to reduce the processing times of the SNN. Future advancements in QRNG platforms can further reduce the costs and size of chips and increase this concept's availability and ease of use.

VIII. Conclusion

This paper proposes a new innovative solution to securing RKE vehicle systems using Machine Learning and Quantum Random Number Generators. SecFob contains many security aspects, each securing the RKE system from various attack vectors. SecFob is the first RKE solution utilizing the Siamese Neural Network algorithm and has proven to be a reliable way to analyze radio frequencies. SecFob can use the SNN to classify incoming transmissions as attacking or legitimate with highly accurate results. SecFob uses QRNG and lightweight AES encryption to secure the keyless entry system handshake process further, making attacks nearly impossible. Furthermore, using timestamping techniques ensures proper signal reception and processing to secure the system against more attacks and, in turn, gives the vehicle owner confidence that their investment is safe from outside hostiles.

IX. Acknowledgment

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